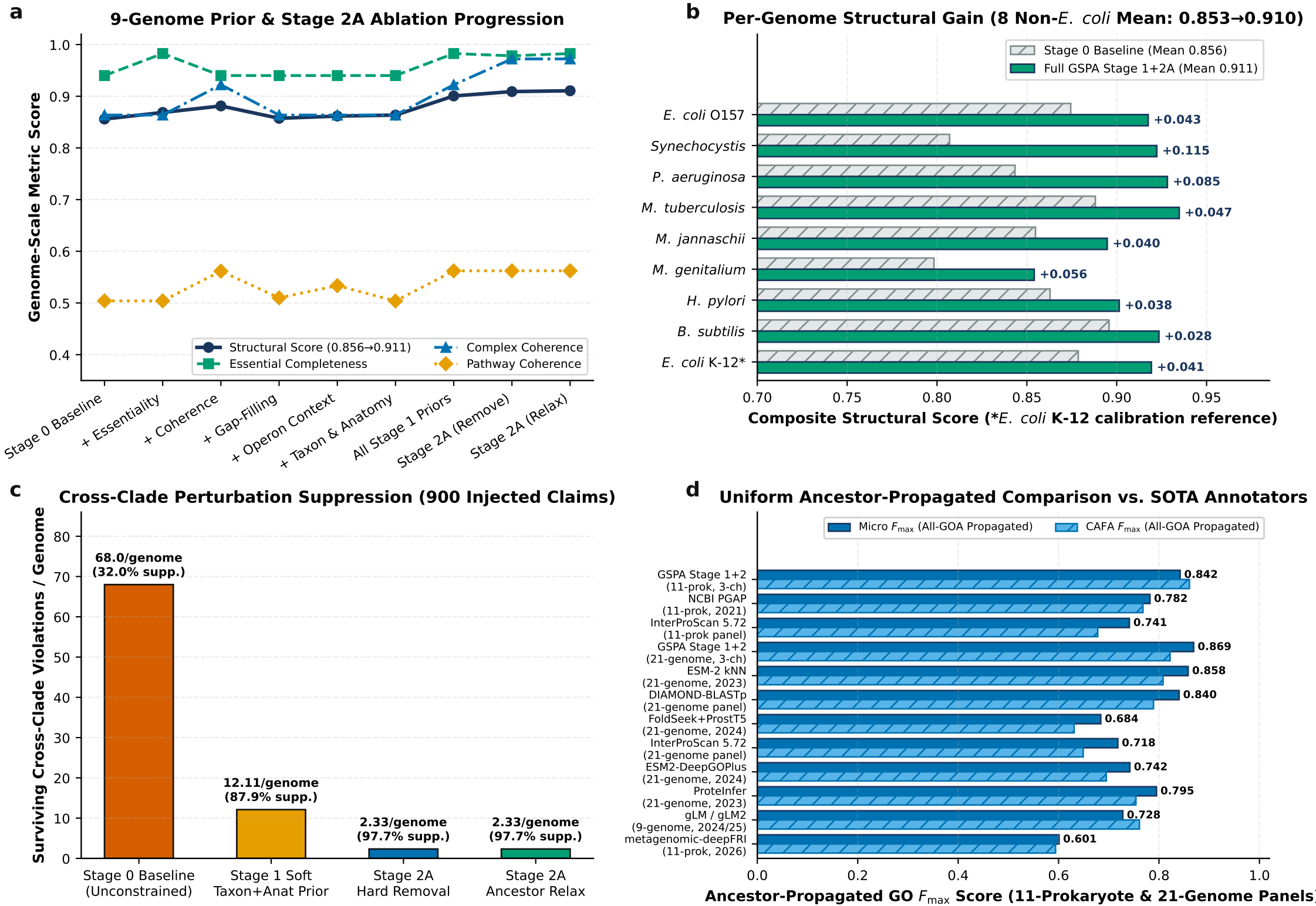

a
9-Genome Prior & Stage 2A Ablation Progression
Genome-Scale Metric Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Structural Score (0.856→0.911)
Essential Completeness
Complex Coherence
Pathway Coherence
Stage 0 Baseline
+ Essentiality
+ Coherence
+ Gap-Filling
+ Operon Context
+ Taxon & Anatomy
All Stage 1 Priors
Stage 2A (Remove)
Stage 2A (Relax)
b
Per-Genome Structural Gain (8 Non-E. coli Mean: 0.853→0.910)
Stage 0 Baseline (Mean 0.856)
Full GSPA Stage 1+2A (Mean 0.911)
E. coli O157
Synechocystis
P. aeruginosa
M. tuberculosis
M. jannaschii
M. genitalium
H. pylori
B. subtilis
E. coli K-12*
+0.043
+0.115
+0.085
+0.047
+0.040
+0.056
+0.038
+0.028
+0.041
0.70
0.75
0.80
0.85
0.90
0.95
Composite Structural Score (*E. coli K-12 calibration reference)
c
Cross-Clade Perturbation Suppression (900 Injected Claims)
Surviving Cross-Clade Violations / Genome
80
70
60
50
40
30
20
10
0
68.0/genome
(32.0% supp.)
12.11/genome
(87.9% supp.)
2.33/genome
(97.7% supp.)
2.33/genome
(97.7% supp.)
Stage 0 Baseline
(Unconstrained)
Stage 1 Soft
Taxon+Anat Prior
Stage 2A
Hard Removal
Stage 2A
Ancestor Relax
d
Uniform Ancestor-Propagated Comparison vs. SOTA Annotators
Micro Fmax (All-GOA Propagated)
CAFA Fmax (All-GOA Propagated)
GSPA Stage 1+2
(11-prok, 3-ch)
NCBI PGAP
(11-prok, 2021)
InterProScan 5.72
(11-prok panel)
GSPA Stage 1+2
(21-genome, 3-ch)
ESM-2 kNN
(21-genome, 2023)
DIAMOND-BLASTp
(21-genome panel)
FoldSeek+ProstT5
(21-genome, 2024)
InterProScan 5.72
(21-genome panel)
ESM2-DeepGOPlus
(21-genome, 2024)
ProteInfer
(21-genome, 2023)
gLM / gLM2
(9-genome, 2024/25)
metagenomic-deepFRI
(11-prok, 2026)
0.842
0.782
0.741
0.869
0.858
0.840
0.684
0.718
0.742
0.795
0.728
0.601
0.0
0.2
0.4
0.6
0.8
1.0
Ancestor-Propagated GO Fmax Score (11-Prokaryote & 21-Genome Panels)